

VVV-QMRF Working Paper v2.0

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When Does a Physical Interaction Become a Valid Registered Measurement?

A VVV-QMRF Registration-Layer Extension of Quantum Mechanics with a Testable Prediction for Extended Wigner's Friend Scenarios

Working Paper v2.0 Author: Viet Nguyen Xuan (VietVunVut) **Repository:** <https://github.com/AlhugART/VVV-QMRF-VietVunVut-Quantum-Measurement-Registration-Framework> **Status:** Working paper submitted for community feedback. All formal claims are Class D (proposed) or Class C (conjecture) unless stated otherwise. Critique is explicitly invited. **Document type:** Publication-facing draft / publication_draft **Date:** 2026-05-17 **Scope:** Registration-layer modeling of Extended Wigner's Friend scenarios within VVV-QMRF. **Out of scope:** Replacement of Standard Quantum Mechanics, revision of the Born rule, consciousness-based collapse, or direct physical detection of K-side structures. **Claim boundary:** Formal notation in this paper defines proposed K-side registration models unless separately identified as Standard Quantum Mechanics or externally validated experimental result. **Formula boundary:** Symbols such as K , $\sigma_R(M)$, requires_K_joint , and $\perp_K$ are registration-layer notation, not new physical observables. **Reader level:** Technical working-paper reader; community feedback is invited before publication-level claims are strengthened. **Neutral wording boundary:** Standard Quantum Mechanics is treated as scope-silent about registration architecture, not as logically defective.

Abstract

The quantum measurement problem has remained unresolved for nearly a century. Standard quantum mechanics specifies, via postulate P3, that a measurement of observable A on state $|\psi\rangle$ yields eigenvalue a_k with probability $|\langle a_k|\psi\rangle|^2$. What P3 does not specify is when a physical interaction constitutes a valid registered measurement event. This silence generates the von Neumann chain problem and leaves the Heisenberg cut formally undefined.

This paper proposes VVV-QMRF (VietVunVut Quantum Measurement Registration Framework): a registration-layer extension of quantum mechanics grounded in Buddhist Pramāṇa epistemology (Dignāga–Dharmakīrti tradition). The framework adds a K-side registration space K , separate from the physical Hilbert space H , and proposes a six-condition test for valid registered measurement. Applied to Extended Wigner’s Friend (EWF) scenarios, this yields a specific conjecture: $K_F \perp K_W$ when a joint registration space K_joint is structurally required but cannot satisfy all six conditions simultaneously.

This conjecture is not trivially true for all frameworks. It separates ordinary Bell-type nonclassicality from LF-level Wigner-friend inconsistency, as illustrated by data from Bong et al. (2020) at $\mu = 0.80, 0.81$ where Bell non-LF violation is reported while Local Friendliness inequalities remain unviolated. Existing data from Proietti et al. (2019) and Bong et al. (2020) are compatible with the prediction under the registration-layer reading developed here. The framework does not replace Standard Quantum Mechanics, revise the Born rule, invoke consciousness, or claim that a K-side structure is directly observed as a physical object. All formal claims are classified by evidence level. A purpose-designed experiment is required for confirmation.

1. The Registration Layer Gap

1.1 What Quantum Mechanics Specifies

Standard quantum mechanics is built on four postulates:

- **P1 (State):** A physical system is represented by a density operator $\rho \in D(H)$.
- **P2 (Observables):** Physical quantities are represented by self-adjoint operators A on H .
- **P3 (Measurement):** Measurement of A on state $|\psi\rangle$ yields eigenvalue a_k with probability $|\langle a_k | \psi \rangle|^2$, after which the state updates to the corresponding eigenstate.
- **P4 (Dynamics):** Between measurements, the state evolves unitarily via the Schrödinger equation.

P3 specifies what outcome appears and with what probability. P3 does not specify when a physical interaction counts as a measurement, what structural conditions distinguish a measurement from a non-measurement interaction, or what stops the measurement chain.

This is the registration layer gap.

1.2 The Von Neumann Chain Problem

Von Neumann (1932) observed that if apparatus $A1$ measures system S , $A1$ becomes entangled with S . If apparatus $A2$ then measures $A1$, $A2$ becomes entangled with $A1+S$. No postulate of standard QM specifies where this chain terminates. Every proposed stopping point — macroscopic size, decoherence, consciousness — is external to the formalism.

Decoherence (Zurek 2003) explains why superpositions become effectively classical at the macroscopic scale. It does not explain when a physical interaction constitutes a registration event in the sense of a definite recorded outcome. The von Neumann chain

problem is a registration-layer problem, not a decoherence problem.

1.3 The Heisenberg Cut

Copenhagen quantum mechanics (Bohr 1935; Heisenberg 1958) assigns measurement authority to “the classical apparatus” without formally defining what makes an apparatus classical, why it has registration authority, or what structural conditions produce a definite outcome. The Heisenberg cut is a boundary without a formal location.

VVV-QMRF addresses both problems by introducing a registration layer K , separate from the physical layer ρ , and proposing formal conditions for valid registration events within K .

2. The K -side Registration Space

2.1 Layer Separation

VVV-QMRF introduces a strict separation between two layers:

- **Physical layer (ρ -side):** Physical states, observables, and dynamics as described by Standard QM. This layer is not modified. $\rho \in D(H)$, H is the standard Hilbert space.
- **Registration layer (K -side):** The space of registration events, recording conditions, and validity states. $K \neq H$. K -side structure does not alter ρ -side dynamics.

This separation is the core architectural commitment of VVV-QMRF.

2.2 Minimal K -state

A registration event is represented by a minimal K -state tuple:

$$k = \langle M, o, \text{cert}, t, V \rangle$$

where:

- M = measurement act identifier
- o = registered outcome
- cert = certification status (self-certified or not)
- t = registration time
- V = validity status $\in \{0, 1\}$

The K -side registration space K is the collection of such tuples produced by a registering system R over time.

2.3 Source: Buddhist Pramāṇa Epistemology

The registration-layer architecture is derived structurally from Buddhist Pramāṇa epistemology (Dignāga–Dharmakīrti tradition), as systematized by Prasad (2023). Pramāṇa theory addresses the formal conditions under which a cognition constitutes a valid knowledge event. The structural extraction used here concerns:

- **Svasaṃvedana:** Self-certifying cognition — a cognition that certifies its own occurrence without requiring a second-order cognition to validate it. Source for E1.
- **Arthakriyā:** Validity as functional success — a cognition is valid if it enables successful engagement with its object. Source for validity conditions in E7.
- **Svataḥ/Parataḥ prāmāṇya:** Initial validity is intrinsic, while invalidity is detected extrinsically. This is used as a structural meta-epistemological principle for E7, not as a standalone Buddhist Epistemology node.
- **Anumāna/Pratyakṣa:** Inference and perception as distinct registration modes. Source for E11, E14.

This is a structural extraction, not a claim that Buddhist epistemology proves quantum mechanics or that quantum mechanics validates Buddhist doctrine.

3. The Valid Registered Measurement Test

This section states the central contribution of the paper: a six-condition test for when a physical interaction constitutes a valid registered measurement.

3.1 The Six Conditions

Interaction X is a valid registered measurement for registering system R if and only if:

Condition 1 (Physical):

X occurs at the physical p-side.

[Standard QM governs this layer. VVV-QMRF does not modify it.]

Condition 2 (Admission):

X is admitted into K-side as measurement act M_X for system R.

[The interaction crosses the layer boundary into K.]

Condition 3 (Process membership):

$M_X \in R$, where $R = \{M_{R1}, M_{R2}, \dots\}$ is the registering process of system R ordered by registration time. [E6]

Condition 4 (Self-certification):

$\sigma_R(M_X) = 1$: M_X self-certifies its occurrence within K_R , determined intrinsically, not by any external $M' \neq M_X$. [E1]

Condition 5 (Default validity):

$V(M_X) = 1$ by default upon admission. [E7]

Condition 6 (Non-invalidity):

No later registration act M' in K_R contradicts M_X such that $V(M_X)$ is revised to $\emptyset$. [E7]

All six conditions must hold for X to count as a valid registered measurement.

3.2 The K-side Stopping Proposition

Proposition (K-side stopping, Class D proposed):

If Conditions 1–6 hold for measurement act M_X of registering system R , then no further K-side registration act is required to certify M_X . The von Neumann registration regress terminates at M_X on the K-side without modifying ρ -side dynamics.

Why this addresses the Heisenberg cut: Condition 4 (self-certification) provides the formal property that Copenhagen assigns to “classical apparatus” without definition. An apparatus has registration authority because $\sigma_R(M) = 1$ is determined intrinsically within K_R , requiring no external certifier. This is the registration-layer account of why the chain stops.

Claim class: D (proposed). Formal proof that Conditions 1–6 are jointly sufficient for chain termination is an open item.

3.3 Observer-Indexed Self-Certification

For multi-observer scenarios, the self-certification function carries an observer index:

$$\sigma_R(M) \in \{0, 1\}$$

$\sigma_R(M) = 1$ iff M occurred as a K-side registration event of registering system R , determined intrinsically within K_R , not by any $M' \neq M$ and not by any $R' \neq R$.

Independence: $\sigma_F(M_F)$ is determined within K_F independently of K_W , and vice versa.

The original $\sigma(M)$ is the single-observer special case: $\sigma(M) = \sigma_R(M)$ where R is unique. All properties of the single-observer case are preserved. [E01 Section 11.2, Class D]

4. Extended Wigner’s Friend and K-side Incommensurability

4.1 The Extended Wigner’s Friend Setup

In the Extended Wigner’s Friend scenario (Frauchiger and Renner 2018; Brukner 2018), two observer pairs interact with a shared quantum system:

- **Friend F** measures quantum system S inside a sealed laboratory. From F’s perspective, a definite outcome o_F is recorded: $\sigma_F(M_F) = 1$.
- **Wigner W** models the entire laboratory ($F + S$) as a unitarily evolving quantum state. From W’s perspective, $F+S$ is in superposition before W’s measurement. W then performs an interference measurement M_W on F’s laboratory.

Standard QM supplies formal descriptions for both perspectives without providing a registration-layer account of their structural incompatibility. This is where the registration layer offers new structure.

4.2 Applying the Six Conditions to Each Observer

Friend F:

$X_F \rightarrow M_F \in R_F$ [Condition 3, E6]
 $\sigma_F(M_F) = 1$ [Condition 4, E1]
 $V_F(M_F) = 1$ [Condition 5, E7]
No M' contradicts M_F [Condition 6, E7]
 $\rightarrow M_F$ is a valid registered measurement within K_F .

Wigner W:

$X_W \rightarrow M_W \in R_W$ [Condition 3, E6]
 $\sigma_W(M_W) = 1$ [Condition 4, E1]
 $V_W(M_W) = 1$ [Condition 5, E7]
No M' contradicts M_W [Condition 6, E7]
 $\rightarrow M_W$ is a valid registered measurement within K_W .

Both observers satisfy the six conditions independently within their own K -sides.

4.3 The requires_K_joint Predicate

The question is whether a joint registration space K_{joint} can contain both K_F and K_W as jointly valid entries.

$\text{requires_K_joint}(F, W, M_F, M_W) = 1$
iff a single K_{joint} is structurally required to contain both
 $k_F = \langle M_F, o_F, \text{cert}_F, t_F, V_F \rangle$ and
 $k_W = \langle M_W, o_W, \text{cert}_W, t_W, V_W \rangle$
as jointly valid entries satisfying Conditions 1-6 for both.

$\text{requires_K_joint}(F, W, M_F, M_W) = 0$
iff K_F and K_W can remain independent K -side spaces without
any joint validity check between them.

Sufficient conditions for $\text{requires_K_joint} = 1$:

- **Condition A (Wigner interference):** W performs an interference measurement on the laboratory containing F+S. M_W registers a superposition description of F+S; M_F registers a definite outcome o_F of the same system S. Both claim K -side validity on the same physical event. $\rightarrow \text{requires_K_joint} = 1$.
- **Condition B (Direct comparison):** F and W directly compare registration records and a logical contradiction is detectable. $\rightarrow \text{requires_K_joint} = 1$.

Sufficient conditions for $\text{requires_K_joint} = 0$:

- **Condition C (No interference, no comparison):** W does not perform interference measurement on F's laboratory. K_F and K_W remain causally isolated. $\rightarrow \text{requires_K_joint} = 0$.
- **Condition D (Separable state):** The shared quantum state $|\psi\rangle$ is separable and M_F , M_W act on non-overlapping subsystems. $\rightarrow \text{requires_K_joint} = 0$.

Claim class: D (proposed). Conditions A–D are sufficient conditions only. Necessary and sufficient characterization of requires_K_joint is an open item.

4.4 Definition Boundary for K_joint and K-side Conflict

Before stating the K_joint failure proposition, the paper uses the following bounded definitions:

K_joint

A proposed registration-layer model that would contain entries from K_F and K_W in one joint validity structure.

$\text{M_W} \perp_{\text{K}} \text{M_F}$

A proposed K-side conflict relation: M_W and M_F cannot both retain validity value 1 inside the same K_joint because their registered contents impose incompatible joint-validity commitments on the same physical event domain.

joint-validity failure

The failure of any proposed K_joint to satisfy Conditions 1-6 for both M_F and M_W while preserving $V(\text{M_F})=1$ and $V(\text{M_W})=1$.

These definitions are registration-layer definitions only. They do not introduce a new physical observable, alter Hilbert-space dynamics, or claim that K_joint is directly detected as a physical object. The relation $\perp_{\text{K}}$ is a Class D proposed formal relation; its use in the empirical prediction remains Class C until a full mathematical characterization and purpose-designed test are supplied.

4.5 K_joint Failure and K-side Incommensurability

Proposition (K_joint failure, Class D proposed):

If $\text{requires_K_joint} = 1$ via Condition A, then no K_joint exists such that: - (i) $\sigma_{\text{F}}(\text{M_F}) = 1$ holds within K_joint [Condition 4 for F] - (ii) $\sigma_{\text{W}}(\text{M_W}) = 1$ holds within K_joint [Condition 4 for W] - (iii) $V(\text{M_F}) = 1$ and $V(\text{M_W}) = 1$ simultaneously [Condition 5+6 for both]

Proof sketch:

Step 1 [E6]: F and W are distinct registering processes R_F and R_W with distinct K-side time sequences.

Step 2 [E1, observer-indexed]: $\sigma_{\text{F}}(\text{M_F}) = 1$ is intrinsic to K_F .
 $\sigma_{\text{W}}(\text{M_W}) = 1$ is intrinsic to K_W . Both hold independently.

Step 3 [Condition A]: M_W registers F+S as superposition (no definite o_F).
 M_F registers o_F as definite outcome. These are contradicting registrations: $\text{M_W} \perp_{\text{K}} \text{M_F}$ – they assert incompatible K-side validity claims on the same physical event.

Step 4 [E7 Axiom 2]: If M' contradicts M , $V(\text{M})$ and $V(\text{M'})$ cannot both equal 1 in the same K-space. $\text{M_W} \perp_{\text{K}} \text{M_F}$ implies $V(\text{M_F}) = 1$ and

$V(M_W) = 1$ cannot simultaneously hold in any K_{joint} .
Therefore K_{joint} satisfying (i)–(iii) does not exist.

K-side incommensurability:

$K_F \perp_K K_W$

iff $\text{requires_K_joint} = 1$ AND no K_{joint} satisfying (i)–(iii) exists.

Claim class of Step 4: D (proposed). Depends on E7 Axiom 2, which is itself Class D.
Full formal proof is an open item listed in Section 7.

4.6 The Falsifiable Prediction

VVV-QMRF predicts a two-part structure:

In Extended Wigner’s Friend experiments, configurations satisfying $\text{requires_K_joint} = 1$ are the configurations in which K-side joint-validity failure is structurally relevant. A violation of Local Friendliness inequalities is expected only when this structural condition is combined with sufficient empirical strength, such as adequate entanglement, visibility, and noise control. Configurations satisfying $\text{requires_K_joint} = 0$ are predicted to produce no LF violation from the K-side registration perspective.

This prediction is falsified if: An EWF experiment produces results better modeled by a single joint-registration structure than by K-side incommensurability in a configuration where $\text{requires_K_joint} = 1$ — i.e., if the experimental data are consistent with one K_{joint} satisfying Conditions 1-6 for both observers simultaneously, under Condition A and under empirical conditions strong enough for LF-level failure to be observable. The empirical absence of LF violation in a weak-entanglement or high-noise regime does not by itself falsify the framework, because Section 5.1 treats $\text{requires_K_joint} = 1$ as structurally necessary but not empirically sufficient for visible LF violation. If a single joint-registration model is empirically favored under the relevant structural and empirical conditions, E7 Axiom 2 or the proposed $\perp_K$ relation requires revision.

Claim class: C (conjecture). Pending experimental operationalization and verification.

5. Experimental Contact

5.1 What Distinguishes VVV-QMRF from Existing Frameworks

The key asymmetry is this: $\text{requires_K_joint} = 1$ is a structurally necessary but not empirically sufficient condition for LF violation. The quantum state must also be sufficiently entangled, and the experiment must have sufficient visibility and noise control, for K_{joint} failure to produce an observable inequality violation. This generates a two-regime structure:

- **Regime 1:** $\text{requires_K_joint} = 1$ AND state weakly entangled $\rightarrow$ Condition A structurally active, K_{joint} structurally required, but empirical LF violation may be absent.
- **Regime 2:** $\text{requires_K_joint} = 1$ AND state strongly entangled $\rightarrow K_{\text{joint}}$ failure is predicted to become visible as LF violation.

A decoherence-only framework does not distinguish these two regimes at the registration-layer level, because it contains no predicate equivalent to `requires_K_joint` for separating Bell non-LF violation from LF-level joint-validity failure.

This is not trivially true for all frameworks. It generates an observable conjecture: Bell non-LF violation can appear in Regime 1 while LF inequalities remain unviolated, and LF violation appears in Regime 2 when entanglement, visibility, and noise conditions are sufficient.

5.2 Proietti et al. (2019)

Reference: Proietti, M. et al. Experimental test of local observer independence. *Science Advances* 5(9), eaaw9832 (2019). DOI: 10.1126/sciadv.aaw9832.

Setup: Six-photon optical experiment with four observers (Alice’s Friend, Bob’s Friend, Alice, Bob). Two measurement choices per external observer:

- A0 / B0: read Friend’s memory record (no interference).
- A1 / B1: Wigner-type joint measurement via nonclassical interference on a 50/50 beam splitter.

Applying `requires_K_joint`:

Term	Configuration	W interference?	Condition	<code>requires_K_joint</code>	Role in S
$\langle A1B1 \rangle$	Both Wigner-type	Both sides	A	1	Added
$\langle A1B0 \rangle$	Alice Wigner, Bob record	Alice side	A	1	Added
$\langle A0B1 \rangle$	Alice record, Bob Wigner	Bob side	A	1	Added
$\langle A0B0 \rangle$	Both record readout	None	C	0	Subtracted

Experimental values (alternative observable protocol):

$$\begin{aligned}
 S &= \langle A1B1 \rangle + \langle A1B0 \rangle + \langle A0B1 \rangle - \langle A0B0 \rangle \\
 S &= 0.571 + 0.577 + 0.573 - 0.662 \\
 S &= 2.407 \quad (\text{reported: } 2.407 \pm 0.073, \text{ violation} > 5\sigma)
 \end{aligned}$$

VVV-QMRF reading: The three terms with `requires_K_joint` = 1 contribute positively to the violated aggregate expression. The one term with `requires_K_joint` = 0 is subtracted. This aggregate sign pattern is consistent with the $K_F \perp_K K_W$ prediction, but it is not a per-configuration confirmation because S is evaluated as a combined inequality expression.

Result: Consistent at Bell-Wigner term-contribution level. Not a per-configuration confirmation; S is an aggregate expression.

5.3 Bong et al. (2020)

Reference: Bong, K.W. et al. A strong no-go theorem on the Wigner’s friend paradox. Nature Physics 16, 1199–1205 (2020). DOI: 10.1038/s41567-020-0990-x. arXiv: 1907.05607.

Setup: Extended Wigner’s Friend scenario with four observers (Charlie, Debbie, Alice, Bob). Three measurement settings per external observer. Friends implemented as photon paths within beam-displacer interferometers. State family:

$$\rho_{\mu} = \mu |\Phi^{-}\rangle\langle\Phi^{-}| + (1-\mu)/2 (|HV\rangle\langle HV| + |VH\rangle\langle VH|)$$

where $\mu \in [0,1]$ controls singlet fraction.

Applying `requires_K_joint` to settings:

Setting pair	Alice side	Bob side	W interference?	Condition	requires_K_jo
x=1, y=1	Ask Charlie	Ask Debbie	None	C	0
x=1, y=2/3	Ask Charlie	Superobserver	Bob side	A	1
x=2/3, y=1	Superobserver	Ask Debbie	Alice side	A	1
x=2/3, y=2/3	Superobserver	Superobserver	Both sides	A	1

μ -threshold behavior and VVV-QMRF reading:

μ regime	Reported result	VVV-QMRF reading
Low μ	No inequalities violated	requires_K_joint = 1 structurally active in x=2/3 settings; LF violation absent because state too weakly entangled
$\mu = 0.80, 0.81$	Bell non-LF violated; LF inequalities NOT violated	Regime 1: Condition A active, K_joint required, but entanglement insufficient for LF-level failure
$\mu \approx 0.87$	First LF violation (Semi-Brukner)	Transition to Regime 2: K_joint failure is interpreted as becoming empirically visible
High μ	All categories violated including Genuine LF	Regime 2: x=2/3, y=2/3 Condition A strongly active, K_joint failure fully exposed

The asymmetric prediction: At $\mu = 0.80, 0.81$, Bell non-LF violation appears while LF inequalities remain unviolated. VVV-QMRF offers a registration-layer account of this asymmetry via the two-regime structure: `requires_K_joint = 1` in superobserver settings (Condition A), but the quantum state at this μ value is not sufficiently entangled for

K_{joint} failure to produce LF-level violation. A framework without a $\text{requires}_{K_{\text{joint}}}$ predicate has no structural account, at the registration-layer level, of why Bell-type violation and LF-type violation may appear at different μ thresholds.

Result: Consistent at structural setting-level and μ -threshold level.

5.4 Overall Condition 3 Status

Both Proietti et al. (2019) and Bong et al. (2020) are compatible with the VVV-QMRF prediction under the registration-layer reading developed here. This is an existing-data compatibility check, not confirmation. A purpose-designed experiment varying $\text{requires}_{K_{\text{joint}}}$ directly as an independent variable is required for confirmation-level evidence.

5.5 Claim-Control Summary

Claim	Current status	Boundary condition
Six-condition registered measurement test	Class D proposed definition	Defines K-side registration validity; does not revise P1-P4 or the Born rule
K-side stopping proposition	Class D proposed proposition	Requires formal proof of joint sufficiency of Conditions 1-6
K_{joint} failure proposition	Class D proposed proposition	Depends on E7 Axiom 2 and the proposed definition boundary for $\perp_K$; full mathematical characterization remains open
$\text{requires}_{K_{\text{joint}}}$ predicate	Class D proposed predicate	Sufficient conditions A-D only; necessary and sufficient characterization remains open
LF-violation prediction	Class C conjecture	$\text{requires}_{K_{\text{joint}}} = 1$ is structurally necessary but not empirically sufficient; entanglement, visibility, and noise control also matter
Proietti/Bong comparison	Existing-data compatibility check	Aggregate-level and setting-level consistency only; not confirmation

This table functions as the claim-control layer for the working paper: formal language is limited by the stated evidence class, and empirical language is limited to compatibility unless a purpose-designed experiment directly tests $\text{requires}_{K_{\text{joint}}}$.

6. Positioning Against Existing Interpretations

Interpretation	Response to EWF paradox	VVV-QMRF difference
Copenhagen	WF is ill-posed; classical apparatus required	E1/E7 formalize why apparatus has registration

Interpretation	Response to EWF paradox	VVV-QMRF difference
Many-Worlds	All outcomes occur; no paradox	authority: $\sigma_{R(M)} = 1$ intrinsically Registration events are singular per K-side, not globally branching
QBism	Facts are agent-relative; no paradox	Agrees on agent-relativity; adds formal K-side structure via six conditions
Relational QM	Facts are relation-relative; no paradox	Closest existing framework; VVV-QMRF adds formal registration machinery that Relational QM does not provide
Objective Collapse	Physical collapse restores absolute facts	VVV-QMRF does not require physical collapse mechanism on p-side

Relation to Relational Quantum Mechanics: Rovelli (1996) argues that quantum states are relative to observers. VVV-QMRF adds the formal registration-layer structure that explains why they are relative: $\sigma_{R(M)}$ operates intrinsically within K_R , independently of $K_{\{R'\}}$. $K_F \perp_K K_W$ is the registration-layer account of Rovelli’s observer-relativity. This is a genuine extension, not a restatement: VVV-QMRF supplies the formal conditions (E1, E6, E7) and the requires_K_joint predicate that Relational QM does not.

7. Scope, Limitations, and Open Items

7.1 What This Paper Does Not Claim

- Does not replace Standard Quantum Mechanics or revise P1–P4.
- Does not revise the Born rule or unitary evolution.
- Does not claim Buddhist epistemology proves quantum mechanics.
- Does not claim K-side incommensurability is experimentally confirmed.
- Does not claim that consciousness plays any role in registration.
- Does not claim the EWF paradox is fully resolved on the p-side.
- Does not claim requires_K_joint is a necessary and sufficient condition (sufficient only).

7.2 Open Items

Open item	Location	Status
Formal proof of Step 4 (E7 Axiom 2 in K_{joint} failure proposition)	Section 4.5	Incomplete
Formal definition of $\perp_K$ as a mathematical relation	Section 4.4	Proposed definition boundary only; full formalism incomplete

Open item	Location	Status
Necessary and sufficient characterization of requires_K_joint	Section 4.3	Sufficient conditions only
Prove equivalence of $\sigma(M)$ and $\hat{R}_{svasa}$ formalisms	E01 Section 11.5	Not started
Axiomatize K as a full mathematical structure (topology, order)	Section 2.2	Partial
Apply requires_K_joint to Proietti raw count-level data	Section 5.2	Aggregate level only
Apply requires_K_joint to individual LF facet terms in Bong	Section 5.3	Setting-level only
Explicit separability verification for Bong ρ_μ regimes	Section 5.3	Not done
Purpose-designed VVV-QMRF experiment	Section 5.4	Not done

7.3 Falsification Condition (Restated)

This paper's central prediction is falsified if:

An Extended Wigner's Friend experiment produces results better modeled by a single joint-registration structure than by K-side incommensurability, for a configuration where requires_K_joint = 1 via Condition A and the empirical regime has sufficient entanglement, visibility, and noise control for LF-level failure to be observable.

In operational terms, this would mean that the data support one registration-layer model in which Conditions 1-6 are jointly satisfiable for both observers with $V(M_F)=1$ and $V(M_W)=1$, rather than a model requiring $K_F \perp_K K_W$. If that occurs under the stated structural and empirical conditions, E7 Axiom 2 or the proposed $\perp_K$ relation requires revision and the K_{joint} failure proposition does not hold.

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All formal claims are Class D (proposed) or Class C (conjecture) Falsification condition
stated in Section 4.6 and Section 7.3 Purpose-designed experiment required for
confirmation